

Orangutan Density Estimation from Nest Surveys

Draft 1 Protocol

Fauna & Flora Indonesia Programme

Jarian Permana, Ryan Avriandy

2026-05-26

Table of contents

0.1	Install and load package	1
1	Approach A — using Distance::ds	2
1.1	Load Dataset	2
1.2	Inspect dataset	2
1.3	Prepare dataset for Distance	3
1.4	Fit half-normal and hazard-rate models	4
1.4.1	Compare models	5
1.5	Extract nest density	6
2	Approach B — using unmarked::distsamp	6
2.1	Create distance bins	6
2.2	Covariate	7
2.3	Create dataset	8
2.4	Fit model	8
2.4.1	Compare models	10
2.5	Extract nest density	10
2.6	Comparison between both approaches	10
3	Convert nest density to orangutan density	11
	References	12

0.1 Install and load package

This analysis uses two different R packages to estimate nest density from line-transect distance sampling data. The first package, ‘Distance’, is used to apply the conventional distance sampling approach, where individual nest detections and their perpendicular distances from the transect line are used to estimate a detection function. The second package, ‘unmarked’, is used to apply a hierarchical distance sampling approach through ‘distsamp()’, where detections are grouped into distance intervals for each transect.

```
library(Distance)
library(unmarked)
library(dplyr)
library(tidyrr)
library(ggplot2)
```

1 Approach A — using Distance::ds

The first approach uses the `ds()` function from the `Distance` package. This follows the standard distance sampling framework commonly used in wildlife surveys. In orangutan nest surveys, the detected objects are not orangutans directly, but orangutan nests. Therefore, the model first estimates nest density by accounting for the fact that nests farther from the transect line are less likely to be detected.

The main objective of this approach is to estimate the detection function and use it to correct the observed number of nests. Without this correction, the density estimate would likely be biased downward because some nests, especially those farther from the transect line or hidden by vegetation, are missed during field observation.

1.1 Load Dataset

```
load("nest_simulated.RData")
```

For a real orangutan nest survey, the minimum required information would include the identity of the **transect**, the **length of each transect**, and the **perpendicular distance** from the transect line to each detected nest. Additional covariates, such as habitat type, forest condition, observer identity, vegetation density, or disturbance level, can also be included if the analysis aims to explain variation in detection probability or nest density.

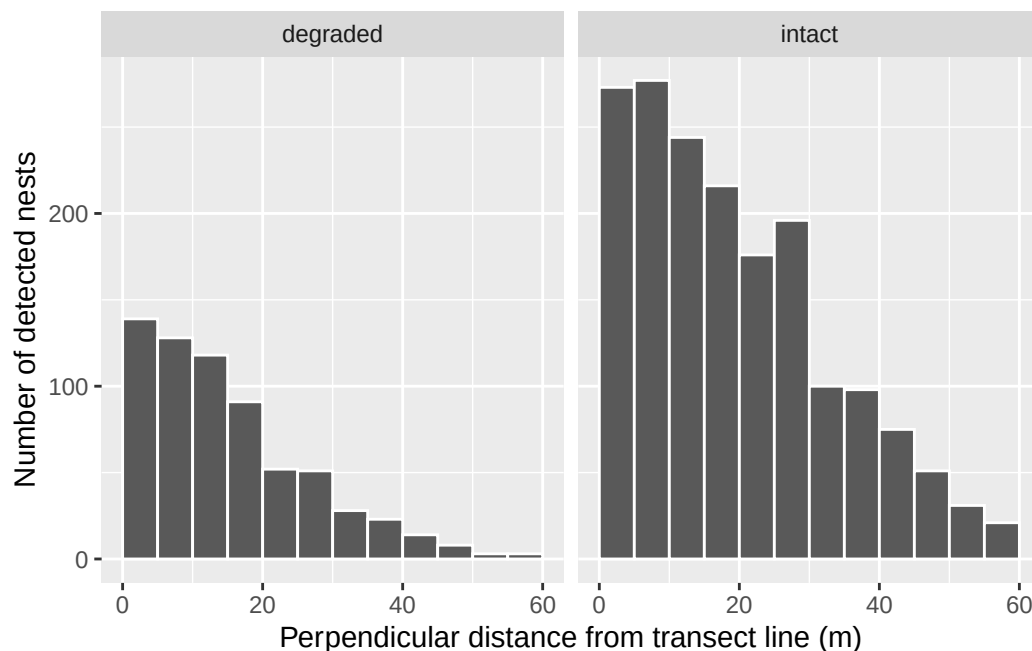
1.2 Inspect dataset

Before fitting a distance sampling model, it is important to inspect the distribution of detected nests across perpendicular distances. The histogram shows how many nests were detected at different distances from the transect line. In a typical distance sampling dataset, most detections are expected to occur close to the transect line, while the number of detections should decline as distance from the line increases.

This declining pattern is important because it provides evidence that detectability decreases with distance. If the histogram does not show this general pattern, the distance sampling model may not fit well, or the survey data may have issues such as inaccurate distance measurement, heaping at rounded values, or inconsistent detection by observers.

In this simulated dataset, the plot also allows us to compare detections between intact and degraded forest. Fewer detections in degraded forest may indicate lower nest density, lower detectability, or a combination of both. At this stage, the plot should be interpreted as exploratory rather than conclusive. The model will later help correct for imperfect detection and provide a formal estimate of nest density.

```
ggplot(nest_obs, aes(x = distance_m)) +  
  geom_histogram(binwidth = 5, boundary = 0, colour = "white") +  
  facet_wrap(~ habitat) + # If you have multiple land cover on transects  
  labs(  
    x = "Perpendicular distance from transect line (m)",  
    y = "Number of detected nests"  
  )
```



The histogram shows that most nests were detected close to the transect line, while fewer nests were detected at greater perpendicular distances. This pattern is expected in distance sampling because detectability usually declines with distance from the observer. The plot also suggests fewer detections in degraded forest compared with intact forest, which may reflect lower simulated nest density, differences in detectability, or both.

1.3 Prepare dataset for Distance

The **Distance** package requires the dataset to follow a specific structure. Each detected object should have a unique object ID, a region label, a sample or transect label, survey effort, and perpendicular distance. In this analysis, each detected nest is treated as one object, each transect is treated as one sample, and the transect length is used as the measure of survey effort.

It is important to ensure that the units are consistent across the dataset. In this example, distance and transect length are converted to kilometres, and the study area is expressed in square kilometres. This allows the resulting density estimate to be interpreted as nests per km². Unit consistency is critical because mismatched units can produce density estimates that are numerically incorrect even if the model runs successfully.

The **Area** value represents the size of the region for which density or abundance is being estimated. In this simulated example, the study area is set to 100 km². In a real survey, this value should represent the actual area of the survey stratum, management block, protected area, or landscape being analysed.

```
distance_data <- nest_obs %>%
  mutate(
    Region.Label = "Study_area",
    Area = 100, # example study area size in km2
    Sample.Label = transect_id,
    Effort = length_km,
    distance = distance_km
  ) %>%
  select(
    object,
    Region.Label,
    Area,
```

```

    Sample.Label,
    Effort,
    distance,
    habitat
  )
head(distance_data)

```

	object	Region.Label	Area	Sample.Label	Effort	distance	habitat
1		1 Study_area	100	T01	2	0.00273339	intact
2		2 Study_area	100	T01	2	0.03168633	intact
3		3 Study_area	100	T01	2	0.02739688	intact
4		4 Study_area	100	T01	2	0.02720005	intact
5		5 Study_area	100	T01	2	0.04065424	intact
6		6 Study_area	100	T01	2	0.03435800	intact

1.4 Fit half-normal and hazard-rate models

Distance sampling models estimate a detection function, which describes how the probability of detecting a nest changes with distance from the transect line. In this analysis, two commonly used detection functions are tested: the half-normal and hazard-rate functions.

The half-normal detection function assumes that detection probability is highest on the transect line and declines smoothly with increasing distance. This function is often appropriate when detectability decreases gradually and the detection histogram shows a clear shoulder near the transect line. The hazard-rate function is more flexible and can sometimes fit data where detectability remains high near the line before dropping more sharply at greater distances.

A truncation distance is also applied. In this example, nests beyond 40 m are excluded from the model. Truncation is commonly used in distance sampling to remove extreme observations that may make the detection function difficult to fit. These far-distance observations can have strong influence on the shape of the detection function, especially when sample size is limited. The truncation distance should be chosen based on the detection histogram, field conditions, and ecological reasoning.

```

m_ds_hn <- ds(
  data = distance_data,
  transect = "line",
  key = "hn",
  adjustment = NULL,
  truncation = 40 / 1000 # look at the plot, we may use 40 m as maximum detection
)

```

Fitting half-normal key function

AIC= -14572.867

```

m_ds_hr <- ds(
  data = distance_data,
  transect = "line",
  key = "hr",
  adjustment = NULL,
  truncation = 40/1000 # look at the plot, we may use 40 m as maximum detection
)

```

Fitting hazard-rate key function

AIC= -14571.68

1.4.1 Compare models

After fitting the half-normal and hazard-rate models, the two models are compared using Akaike's Information Criterion, or AIC. AIC is used to compare alternative models, where the model with the lower AIC is generally considered to have better support from the data, after accounting for model complexity.

In addition to AIC, the detection function plots should also be inspected. A suitable detection function should show high detection probability near the transect line and a plausible decline with increasing distance. For line-transect distance sampling, it is generally expected that detection probability is highest at or near zero distance. A model with a lower AIC but an unrealistic detection curve should be treated cautiously.

In this example, the half-normal model has the smaller AIC and shows a reasonable detection curve with a clear shoulder near the transect line. This suggests that the half-normal model is suitable for estimating nest density from the simulated dataset. Therefore, the half-normal model is selected for the next step.

```
summary(m_ds_hn)
```

```
summary(m_ds_hr)
```

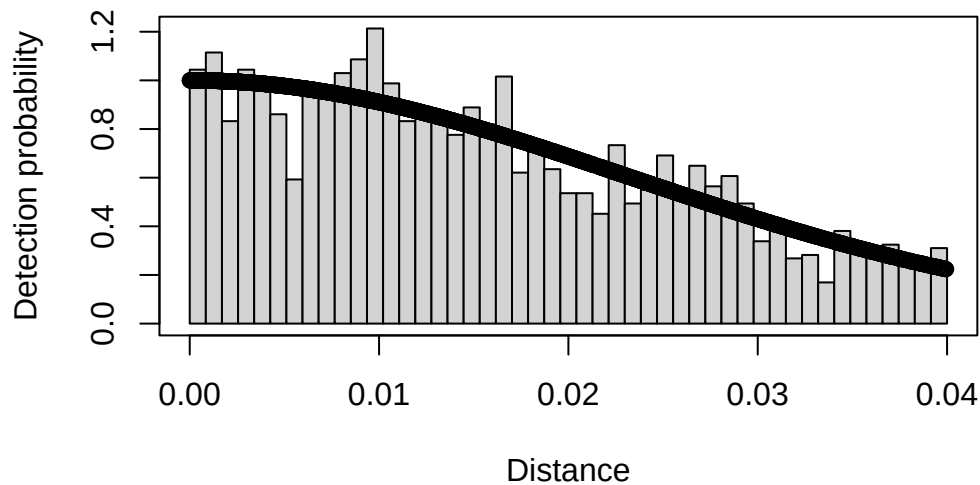
```
AIC(m_ds_hn, m_ds_hr)
```

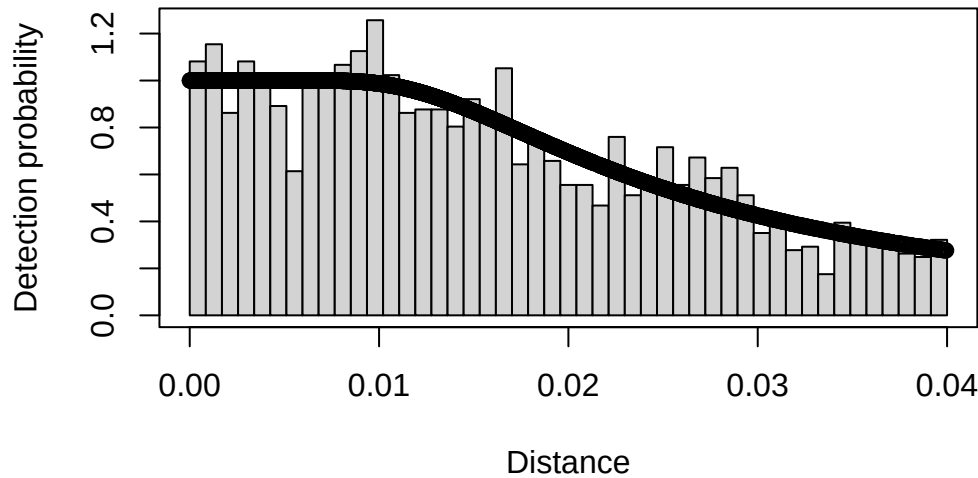
```
plot(m_ds_hn)
```

```
plot(m_ds_hr)
```

```
ds_sum <- summary(m_ds_hn)
```

	df	AIC
m_ds_hn	1	-14572.87
m_ds_hr	2	-14571.68





1.5 Extract nest density

After selecting the preferred detection function, the next step is to extract the estimated nest density from the model output. The `Distance::ds()` model estimates density after correcting the observed number of nests for imperfect detection. This means the resulting value represents the estimated number of nests per km², not simply the raw count of observed nests divided by the surveyed area.

```
# Estimated nest density, nests per km2
ds_sum$dht$individuals$D
```

	Label	Estimate	se	cv	lcl	ucl	df
1	Total	520.5198	38.19805	0.07338444	449.084	603.3189	44.86187

In this example, the half-normal model estimates nest density at approximately 520.52 ± 38.20 nests per km². This estimate should be interpreted as the density of visible orangutan nests in the surveyed landscape, after accounting for the fact that some nests were missed during observation.

However, this is still not orangutan density. Orangutan nest surveys estimate the density of nests first. To estimate orangutan density, nest density must later be converted using biological parameters: the proportion of individuals that build nests, the nest production rate, and the nest decay time.

2 Approach B — using `unmarked::distsamp`

2.1 Create distance bins

The `unmarked::distsamp()` function requires detections to be grouped into distance classes. In this example, the distance intervals are set at 10 m increments from 0 to 60 m. Each detected nest is assigned to one of these distance intervals based on its perpendicular distance from the transect line.

The resulting object is a detection matrix, where each row represents a transect and each column represents a distance interval. The values in the matrix indicate the number of nests detected in each distance class for each transect.

This step is important because `distsamp()` does not use the exact distance to each nest in the same way as

`Distance::ds()`. Instead, it uses the frequency of detections within predefined distance intervals. Therefore, the choice of distance breaks can influence the model. Intervals that are too wide may lose important information, while intervals that are too narrow may create many empty cells and make the model less stable.

```
dist_breaks <- c(0, 10, 20, 30, 40, 50, 60)
```

```
nest_obs_um <- nest_obs %>%
  mutate(
    transect_id = factor(transect_id)
  )
```

```
y <- formatDistData(
  distData = nest_obs_um,
  distCol = "distance_m",
  transectNameCol = "transect_id",
  dist.breaks = dist_breaks
)
```

```
head(y)
```

	[0,10]	(10,20]	(20,30]	(30,40]	(40,50]	(50,60]
T01	26	24	19	12	4	3
T02	30	27	13	9	8	2
T03	28	20	19	11	6	0
T04	24	32	20	11	5	2
T05	28	20	20	9	9	1
T06	26	19	18	13	1	3

2.2 Covariate

Site covariates are prepared to describe each transect. In this simulated dataset, the main covariate is habitat type, which distinguishes between intact and degraded forest. Transect length is also included because line-transect effort must be provided to the `unmarkedFrameDS` object.

In a real orangutan nest survey, this section could include additional covariates that may influence nest density or detection probability. For example, forest type, canopy cover, elevation, distance to river, distance to settlement, or disturbance intensity could be used to model variation in nest density. Observer identity, visibility, slope, or understory density could be used to model variation in detection probability.

At this stage, the covariate table must have one row per transect and must be aligned with the rows of the detection matrix. This alignment is essential. If the order of rows in the covariate table does not match the order of transects in the detection matrix, the model may assign the wrong covariate values to the wrong transects.

```
site_covs <- nest_obs %>%
  mutate(
    habitat = factor(habitat),
    transect_id = factor(transect_id),
    length_m = length_km * 1000
  ) %>%
  select(transect_id, habitat, length_m) %>%
  distinct(transect_id, habitat, length_m)

rownames(site_covs) <- site_covs$transect_id
site_covs <- site_covs[rownames(y), ]
```

```
head(site_covs)
```

	transect_id	habitat	length_m
T01	T01	intact	2000
T02	T02	intact	2000
T03	T03	intact	2000
T04	T04	intact	2000
T05	T05	intact	2000
T06	T06	intact	2000

2.3 Create dataset

The detection matrix and site covariates are combined into an `unmarkedFrameDS` object. This object stores all information required by `distsamp()`, including the count of detected nests in each distance interval, the distance breaks, transect length, survey type, and measurement units.

The survey type is specified as "line" because the simulated survey represents line-transect sampling. The transect length is provided in metres, and the distance breaks are also in metres. The `unitsIn = "m"` argument tells `unmarked` that the input distances are measured in metres.

Creating the `unmarkedFrameDS` object is a key step because it formalises the dataset into the structure required for hierarchical distance sampling. Once this object is created, different models can be fitted by changing the formula, detection function, or covariates.

```
umf <- unmarkedFrameDS(  
  y = y,  
  siteCovs = site_covs,  
  dist.breaks = dist_breaks,  
  tlength = site_covs$length_m,  
  survey = "line",  
  unitsIn = "m"  
)  
  
head(umf)
```

Data frame representation of `unmarkedFrame` object.

	y.1	y.2	y.3	y.4	y.5	y.6	transect_id	habitat	length_m
T01	26	24	19	12	4	3	T01	intact	2000
T02	30	27	13	9	8	2	T02	intact	2000
T03	28	20	19	11	6	0	T03	intact	2000
T04	24	32	20	11	5	2	T04	intact	2000
T05	28	20	20	9	9	1	T05	intact	2000
T06	26	19	18	13	1	3	T06	intact	2000

2.4 Fit model

We can use the same function as previous approach which half-normal and hazard rate

```
hn_um_1 <- distsamp(  
  formula = ~ 1 ~ 1,  
  data = umf,  
  keyfun = "halfnorm",  
  output = "density",  
  unitsOut = "kmsq"  
)
```



```
summary(hn_um_1)
```

Call:

```
distsamp(formula = ~1 ~ 1, data = umf, keyfun = "halfnorm", output = "density",
         unitsOut = "kmsq")
```

Density (log-scale):

Estimate	SE	z	P(> z)
6.25	0.026	241	0

Detection (log-scale):

Estimate	SE	z	P(> z)
3.16	0.0176	180	0

AIC: 1628.922

Number of sites: 40

Survey design: line-transect

Detection function: halfnorm

UnitsIn: m

UnitsOut: kmsq

```
hr_um_1 <- distsamp(
  formula = ~ 1 ~ 1,
  data = umf,
  keyfun = "hazard",
  output = "density",
  unitsOut = "kmsq"
)
```

```
summary(hr_um_1)
```

Call:

```
distsamp(formula = ~1 ~ 1, data = umf, keyfun = "hazard", output = "density",
         unitsOut = "kmsq")
```

Density (log-scale):

Estimate	SE	z	P(> z)
5.53	0.0203	272	0

Detection (log-scale):

Estimate	SE	z	P(> z)
5.27	NaN	NaN	NaN

Hazard-rate(scale) (log-scale):

Estimate	SE	z	P(> z)
0.741	NaN	NaN	NaN

AIC: 2858.58

Number of sites: 40

```
Survey design: line-transect
Detection function: hazard
UnitsIn: m
UnitsOut: kmsq
```

2.4.1 Compare models

The half-normal and hazard-rate models are compared using model selection. Similar to the `Distance` approach, the model with the lower AIC is generally considered to have stronger support from the data. This comparison helps determine which detection function better describes the decline in nest detectability with increasing distance from the transect line.

In this example, the half-normal model is selected because it has stronger support than the hazard-rate model. This is consistent with the result from the `Distance::ds()` approach, where the half-normal detection function also performed better. The consistency between the two approaches suggests that the detection pattern in the simulated dataset is well described by a smooth decline in detectability with distance.

```
mlist <- fitList(hn_um_1, hr_um_1)

modSel(mlist)
```

	nPars	AIC	delta	AICwt	cumltvWt
hn_um_1	2	1628.92	0.00	1.0e+00	1.00
hr_um_1	3	2858.58	1229.66	9.6e-268	1.00

2.5 Extract nest density

After selecting the half-normal model, the state parameter is predicted from the model. Because the model was fitted with `output = "density"` and `unitsOut = "kmsq"`, the state prediction represents nest density in nests per km².

For the current model, the formula `~ 1 ~ 1` assumes constant density across transects. As a result, the predicted density is expected to be the same, or nearly the same, for all transects. Therefore, it is appropriate to extract one representative prediction from the model output.

```
pred_um <- predict(
  hn_um_1,
  type = "state"
)

# Average predicted nest density across transects
D_nest_um <- pred_um %>%
  summarise(Estimate = mean(Predicted),
            SE = mean(SE))
D_nest_um
```

	Estimate	SE
1	517.6366	13.45022

The `unmarked::distsamp()` model estimated nest density at approximately 517.64 ± 13.45 nests per km².

2.6 Comparison between both approaches

Overall, the two approaches produced very similar estimates of nest density. The `Distance::ds()` model estimated nest density at 520.52 ± 38.20 nests per km², while the `unmarked::distsamp()` model estimated nest density at 517.64 ± 13.45 nests per km². This suggests that the simulated dataset contains a clear and consistent detection pattern that can be captured by both modelling frameworks.

For reporting purposes, `Distance::ds()` is often easier to explain because it follows the conventional distance sampling workflow. It is particularly useful when the main objective is to estimate density and report the fitted detection function. The model diagnostics and plots are also helpful for assessing whether the detection function is appropriate.

In contrast, `unmarked::distsamp()` is especially useful when the analysis aims to explore ecological drivers of nest density. For example, it can be used to examine whether nest density varies between intact and degraded forest, or whether detection probability changes with vegetation density or observer identity. Therefore, `Distance::ds()` can be treated as the main density-estimation approach, while `unmarked::distsamp()` can be used as a complementary modelling approach for ecological interpretation.

It is important to note that this estimate is still nest density, not orangutan density. Orangutan density is calculated in the next step by converting nest density using three biological parameters: the proportion of individuals that build nests (p), the nest production rate (r), and the nest decay time (t).

3 Convert nest density to orangutan density

The conversion was like this:

$$D_{orangutan} = \frac{D_{nest}}{p * r * t}$$

p as the proportion of nest builders, r as nests built per individual per day, and t as nest decay time

The logic behind this equation is that the number of visible nests in the forest is influenced by how many orangutans are present, how often each orangutan builds nests, and how long those nests remain visible. If nests remain visible for a long time, many nests can accumulate in the forest even when orangutan density is relatively low. Conversely, if nests decay quickly, the same observed nest density may indicate a higher orangutan density.

Table 1: Parameter from the literature for p, r and t

Parameter	Value	Landscape / Context	Sources
p	0.89 / 0.90	Gunung Palung, West Kalimantan	Johnson et al. (2005)
r	1.16	Gunung Palung, West Kalimantan	Johnson et al. (2005)
t	276	Gunung Palung, West Kalimantan	Johnson et al. (2005)

The selection of nest decay time requires more caution. Published values vary widely across landscapes. Therefore, the recommended approach is to use local nest decay data if available. If local decay monitoring was not conducted, the analysis should present a sensitivity analysis using multiple plausible values. This is more transparent than selecting only one decay value, especially when the objective is to understand how sensitive orangutan density estimates are to assumptions about nest persistence.

For the this context we will try using reference from Gunung Palung and using nest density from `unmarked::distsamp()`:

```
D_nest <- 517.64
SE_D_nest <- 13.45

p <- 0.89
r <- 1.16
```

```

t <- 276

D_orangutan <- D_nest / (p * r * t)

SE_D_orangutan <- SE_D_nest / (p * r * t)

lower_95 <- D_orangutan - 1.96 * SE_D_orangutan
upper_95 <- D_orangutan + 1.96 * SE_D_orangutan

data.frame(
  D_orangutan = D_orangutan,
  SE = SE_D_orangutan,
  lower_95 = lower_95,
  upper_95 = upper_95
)

```

```

  D_orangutan      SE lower_95 upper_95
1    1.816648 0.04720252 1.724131 1.909165

```

Then you can report your orangutan density as 1.81 ± 0.05 individuals per km^2

References

Johnson, Andrea E., Cheryl D. Knott, Bhayu Pamungkas, Monalisa Pasaribu, and Andrew J. Marshall. 2005. "A Survey of the Orangutan (*Pongo Pygmaeus Wurmii*) Population in and Around Gunung Palung National Park, West Kalimantan, Indonesia Based on Nest Counts." *Biological Conservation* 121 (4): 495–507. <https://doi.org/10.1016/j.biocon.2004.06.002>.